AI acknowledgement coversheet

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Your details

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Assessment details

Course: DECO7230 Digital Prototyping and Extended Reality

Assignment: Design Concept Report

AI use

Select the option that applies:

☒ Yes, AI tools **have** been used to generate or edit material in this document

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- AI tool (e.g. Copilot, ChatGPT, Claude, Gemini)
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- prompt(s) used
- relevant part, section, or page of the assessment
- date.

Complete the table below with this information.

Tool	Use	Prompt(s)	Section	Date
ChatGPT	Used to check spelling and grammar and edit my final document	Ensure my report's grammatical accuracy	Entire work	12 August 2026

This coversheet has been adapted with thanks to BIOM3020 Integrated Endocrinology.

MiroXR: Embodied Sticky Note Organisation

Concept Overview

XR reimagines Miro's sticky-note organisation function as an immersive spatial interaction. In desktop Miro, users typically select a sticky note using a mouse cursor, drag it across a two-dimensional canvas, and release it in a desired section. Although efficient, this interaction remains indirect because the user manipulates the note through a mouse or trackpad.
XR transforms this interaction into a direct, hand-based XR experience. The user stands in front of a large virtual brainstorming board containing a collection of sticky notes and several labelled organisational sections. Instead of controlling a cursor, the user interacts with the notes directly using their hand.
The project does not aim to reproduce the entire Miro application in XR. It focuses specifically on the select–move–place interaction and investigates whether organising digital information can become more tangible, intuitive and embodied through spatial hand interaction.

Design Opportunity

The main design opportunity is to reduce the distance between the user's physical action and the digital object being manipulated.
In desktop Miro: Hand → Mouse/Trackpad → Cursor → Sticky Note
In XR: Hand → Sticky Note
By treating sticky notes as spatial objects rather than flat interface elements, the concept explores whether users can organise information in a way that feels closer to physically moving sticky notes on a real brainstorming wall.

XR Environment

The experience takes place in a single-user virtual brainstorming workspace. A large Miro-style board is positioned vertically in front of the user at a comfortable interaction distance.
Sticky notes appear as lightweight virtual objects on the board. The board is divided into labelled areas such as: Unsorted Notes , Problems , Positive Findings , Opportunities
The user organises notes by directly selecting and moving them between these areas.

User Tasks and Goals

The concept focuses on three connected user tasks:
1. Identify and select a sticky note
The user moves their index finger or hand towards a sticky note. When the hand enters its interaction range, the note begins to glow, indicating that it is interactive. When selected, the note slightly increases in size, confirming that it is currently active.
2. Move the selected sticky note
The user maintains the selection and moves their hand through the virtual workspace. The sticky note follows the movement of the hand, creating a direct manipulation relationship between the user and the digital object.
3. Place the sticky note in a target section
The user moves the selected note into the intended category and releases the interaction. The note remains at the new position and returns to its normal size, indicating that the manipulation has finished.This follows the course emphasis on hand-, gesture- and movement-based embodied interaction rather than relying only on conventional interface buttons.

Interaction and Affordances

The interaction flow is:
Approach → Glow → Select → Enlarge → Drag → Release → Place

Visual feedback is used to communicate each interaction state:

State	Feedback	Purpose
Normal	Normal sticky note	Idle state
Hand approaches	Note glows	Shows that the note is interactable
Selected	Note slightly enlarges	Confirms successful selection
Moving	Note follows the hand	Indicates direct manipulation
Released	Returns to normal size	Confirms that placement is complete

The glow and scale changes act as XR affordances. They are intended to help users understand what actions are possible without requiring detailed instructions, consistent with the course examples of highlighting and visual state changes for interactable XR objects.

Preliminary Paper Prototype Evaluation

A low-fidelity paper prototype test was conducted in Week 2 to explore the core grab–move–place interaction. Participants organised paper sticky notes into Problems, Positive Findings, and Opportunities while imagining an XR brainstorming environment. The test used Wizard-of-Oz and Think-Aloud methods to observe how users naturally selected, moved, and placed notes, as well as what spatial layout and feedback they expected. The findings will inform the next digital XR prototype.

Design Assumptions and Next Prototype

The next prototype will test whether users understand that:
a glowing note is selectable;
a slightly enlarged note is selected;
moving the hand drags the note; and
releasing the interaction places the note.
The next XR prototype will therefore implement:
Approach → Glow → Select → Enlarge → Drag → Place → Release
The focus is to refine the clarity and intuitiveness of the core interaction rather than add additional Miro functions.

Storyboard

The storyboard illustrates the core select–move–place interaction in XR, showing how users organise virtual sticky notes through direct hand interaction. It also demonstrates the visual feedback used at each stage, including glow on approach and enlargement on selection, to make the interaction clear and intuitive.

